

Yeshwanth Soma

Austin, Texas, United States

Phone: 813-539-8834 | Email: soma.yeshwanth1@gmail.com

LinkedIn: linkedin.com/in/soma-yeshwanth

PROFESSIONAL SUMMARY

Solution-Driven Staff Software Engineer with 5+ years of professional development experience and Master's degree in Business Analytics. Expert in architecting enterprise-scale applications with advanced AI/ML automation, microservices architecture, and cloud infrastructure. Proven track record of reducing operational overhead by 90%, processing 25K+ transactions per minute, and leading cross-functional teams of 200+ engineers. Specialized in Java Spring Boot, distributed systems, and intelligent automation solutions.

PROFESSIONAL EXPERIENCE

VISA INC - Austin, TX

Staff Software Engineer

August 2022 - Present

- Architected enterprise-wide AI automation infrastructure deploying **MCP server ecosystem (Jira, GitHub, Bitbucket, Artifactory, SonarQube)** with CLI integration, reducing developer onboarding effort by 50% across 200+ engineers
- Engineered intelligent Security Fix Agent using **LangGraph** and **LangChain** frameworks with **AI automation** to fetch vulnerabilities, update dependencies, and generate pull requests, reducing security patch time from days to under 1 hour across 80+ repositories with 90% reduction in manual overhead
- Developed advanced Sanitization Agent using **OpenAI LLMs** and **ChromaDB vector search** to identify and redact PII from Jira tickets, reducing manual review time by 75% while maintaining 98% accuracy
- Designed **RAG-based** story generation system leveraging **code embeddings** and **LLMs** for context-aware Jira tickets, improving grooming efficiency by 60% and story quality consistency
- Built enterprise email processing application using **Java Spring Boot** with **RAG techniques** for payment authorization and reconciliation, achieving 95% parsing accuracy and 25% faster processing in high-volume environments with **React-based** dashboard
- Created and maintained multiple internal SDKs including **O365 operations SDK**, standardizing integrations and accelerating development across 15+ teams with **React-based** configuration interfaces
- Implemented scalable payment platform using **Java Spring Boot** microservices, **Kafka**, and **Temporal**, processing 25K transactions per minute with 99.9% accuracy and 35% latency reduction
- Enhanced database scalability implementing **PgBouncer** connection pooling and **Kafka Streams**, reducing database load by 50% while maintaining ACID compliance
- Implemented active-active multi-regional architecture using **Azure Load Balancer** and **Traffic Manager** with geo-routing and failover, achieving near-zero downtime across data centers
- Optimized performance with distributed caching integrating **Hazelcast** (45% latency reduction) and **PostgreSQL BDR** for multi-regional failover, achieving 99.99% data availability
- Streamlined deployment operations with **Docker** and **Kubernetes**, **Jenkins** and **Maven** CI/CD pipelines, and blue-green deployments ensuring 99.9% service availability
- Orchestrated automated workflows using **Control-M** across microservices, reducing manual intervention by 75%
- Engineered end-to-end AI-powered Development Automation System using Claude Agent SDK and multi-agent orchestration to automate the full software development lifecycle from Low-Level Design (LLD) to Git Pull Request reducing feature development time by 70% and eliminating manual handoffs

- Established monitoring framework with **Prometheus**, **Grafana**, and **Alertmanager**, reducing MTTR by 40% and improving troubleshooting efficiency by 50% via **Splunk** integration and custom **React** dashboards
 - Implemented security practices using **JMeter** load testing, **SonarQube** and **Checkmarx** analysis, and **NexusIQ** dependency management, reducing vulnerabilities by 85%
-

BLACKROCK - Atlanta, GA

Aladdin Client Services Summer Analyst

June 2022 - August 2022

- Delivered comprehensive **Aladdin** product demonstrations using **SQL** and **UNIX**, improving client understanding and platform adoption
 - Developed product enhancement proposal streamlining client inquiries, increasing employee productivity and client satisfaction through optimized support workflows
 - Conducted stakeholder analysis through interviews and surveys, gathering insights that guided product development and strategic decision-making
-

NCR CORP - Hyderabad, India

Professional Services Software Engineer

July 2019 - December 2020

- Developed end-to-end **POS application** managing front-end and middleware components for enterprise retail solutions
- Delivered high-impact POS solution using **C#**, **HTML**, and **MS SQL**, achieving 80% success rate in resolving critical issues and maintaining uptime
- Accelerated development by 60% through comprehensive documentation, standardized practices, and proactive client requirement management
- Designed fault-tolerant database schema enhancing reliability and performance and ensuring data integrity in high-volume transaction processing

KEY PROJECTS

Debunker - AI-Powered Multi-Modal Misinformation Combat Platform

*Full-Stack Web Application | **FastAPI**, **React**, **PostgreSQL**, **OpenAI GPT-4**, **Google Fact Check Tools**, **Anthropic Claude**, **Grok Live Search**, Audio Processing*

Live Platform: www.de-bunker.com

- Architected production-grade multi-modal fact-checking platform processing both text claims and audio content with 78% accuracy, integrating 4 major AI APIs (**OpenAI GPT-4**, **Google Fact Check Tools**, **Anthropic Claude**, **Grok Live Search**) for comprehensive real-time verification
- Engineered advanced audio intelligence pipeline using **OpenAI Whisper** for speech-to-text transcription with **librosa**-based prosody analysis, implementing acoustic feature extraction (pitch variation, energy levels, tempo) to detect sarcasm probability, emotional context, and potential misinformation markers in spoken content
- Built professional multi-source intelligence synthesis system with weighted confidence scoring algorithms, scientific consensus detection with automatic boosting for peer-reviewed sources, and intelligent result aggregation from authoritative fact-checking databases, achieving sub-2 second response times for complex claims
- Developed real-time social media intelligence module via **X/Twitter Grok Live Search** integration, tracking viral misinformation patterns, community consensus analysis, engagement metrics, and automated trending

falsehood discovery with controversy scoring algorithms

- Implemented enterprise-scale automated content discovery system with multi-source news aggregation (**Reddit API**, **NewsAPI**, social platforms), AI-powered claim extraction using **GPT-4** for identifying fact-checkable statements, and intelligent category classification across health, politics, science, and technology domains
- Created production-ready asynchronous architecture with **FastAPI** async/await patterns throughout, background automation for scheduled claim discovery and fact-checking, comprehensive monitoring with detailed logging and health checks, and **Railway** cloud deployment with **PostgreSQL** for high availability
- Established enterprise security and reliability features including **HTTP Basic Authentication** for protected endpoints, rate limiting (30 requests/minute) with graceful degradation, **Pydantic** model validation with file size restrictions, and intelligent fallback mechanisms when external APIs are unavailable
- Designed dual-input responsive frontend with **React** featuring drag-and-drop audio upload (mp3, wav, m4a support), real-time processing indicators, waveform visualization, structured result presentation with confidence scores, and mobile-responsive design for cross-platform accessibility
- Achieved exceptional performance metrics: 95% uptime on **Railway** cloud platform, processing 1000+ daily fact-check requests, sub-2 second average response time for text claims, 15-second average for audio analysis with transcription, supporting files up to 25MB with optimized audio processing libraries

EDUCATION

Master of Science in Business Analytics and Information Systems

University of South Florida | GPA: 3.6/4.0 | August 2022

Bachelor of Technology in Computer Science

VNRVJIET | GPA: 3.6/4.0 | May 2019

TECHNICAL SKILLS

Programming Languages: Java, Python, C#, C++, JavaScript, TypeScript, HTML, CSS, SQL, NodeJS

AI/ML and Data Science: OpenAI API, Mistral, Claude, Google Gemini, RAG (Retrieval Augmented Generation), Vector Databases, ChromaDB, Semantic Search, LLM Integration, NLP, NLTK, Hugging Face Transformers, Embeddings, AI Automation, LangGraph, LangChain

Databases: PostgreSQL, Redis, MySQL, Oracle SQL, MongoDB, Elasticsearch, MS SQL, PgBouncer, Database Connection Pooling

Frameworks and Tools: Spring Boot, Spring Framework, React, Angular, FastAPI, Kafka, Temporal, Control-M, Hazelcast, Maven, Git, Jira, Jenkins, Docker, Kubernetes, Apache Tomcat, REST API, GraphQL

Cloud and Infrastructure: AWS (Lambda, EC2, S3, CloudWatch), Azure (Load Balancer, Traffic Manager), Google Cloud Platform (GCP), Railway, Multi-Regional Architecture, CI/CD, Infrastructure as Code, Blue-Green Deployments

Monitoring and Security: Prometheus, Grafana, Splunk, SonarQube, Checkmarx, NexusIQ, JMeter, Cucumber, JUnit, Alertmanager, Application Performance Monitoring

Leadership and Soft Skills: Technical Leadership, Cross-functional Collaboration, Architecture Design, Performance Optimization, Documentation, Stakeholder Management, Agile Methodologies, Scrum